Automata and Formal Languages

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Mealy Machine

- In a Mealy machine, the output produced by each transition depends on the internal state prior to the transition and the input symbol used in the transition.

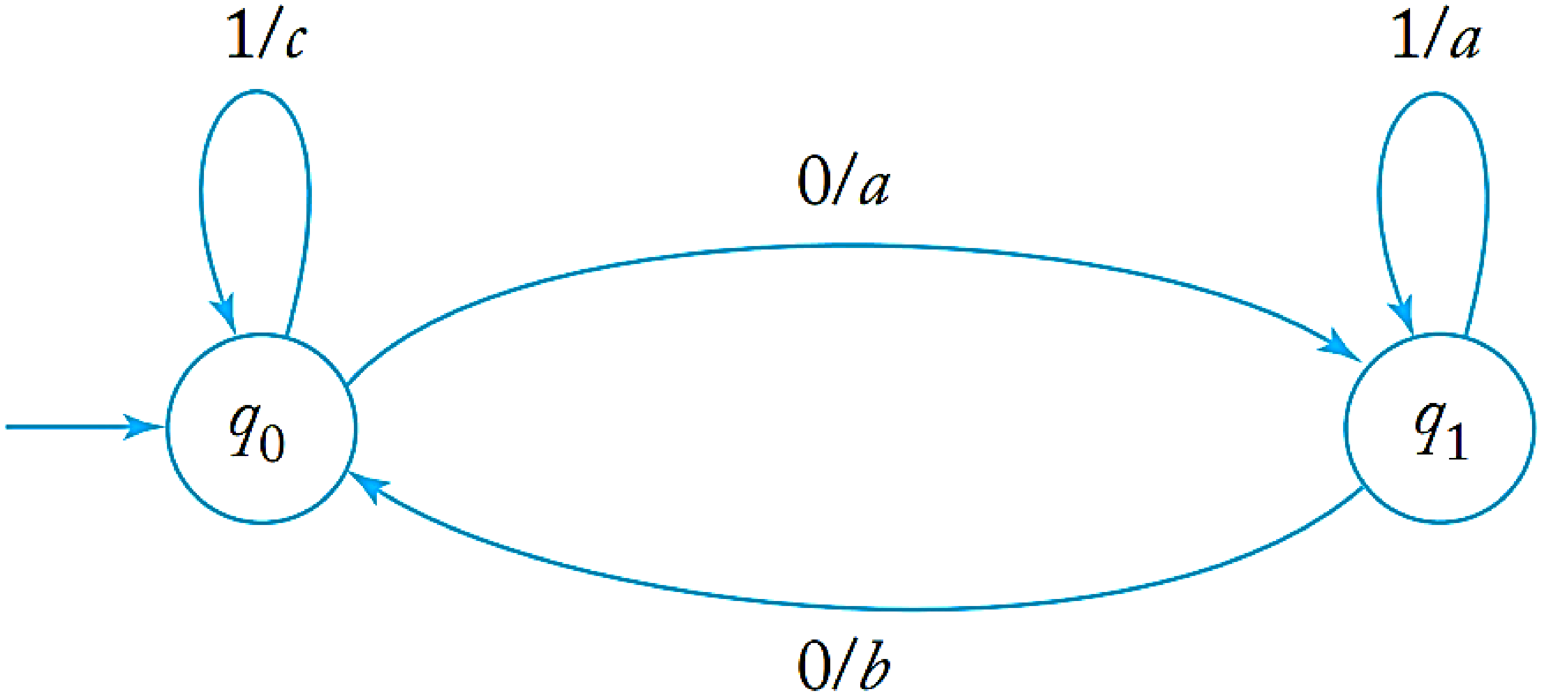
- A Mealy machine is defined by the six tuple $M = (Q, \Sigma, \Gamma, \delta, \theta, q_0)$

- where
  - Q is a finite set of internal states,
  - Σ is the input alphabet,
  - Γ is the output alphabet,
  - $\delta : Q \times \Sigma \to Q$ is the transition function,
  - $\theta : Q \times \Sigma \to \Gamma$ is the output function,
  - $q_0 \in Q$ is the initial state of M.

Mealy Machine

- The mealy machine with $Q = \{q_0, q_1\}, \Sigma = \{0, 1\}, \Gamma = \{a, b, c\}$, initial state q_0, and

- $\delta(q_0, 0) = q_1$,
- $\delta(q_0, 1) = q_0$,
- $\delta(q_1, 0) = q_0$,
- $\delta(q_1, 1) = q_1$,
- $\theta(q_0, 0) = a$,
- $\theta(q_0, 1) = c$,
- $\theta(q_1, 0) = b$,
- $\theta(q_1, 1) = a$



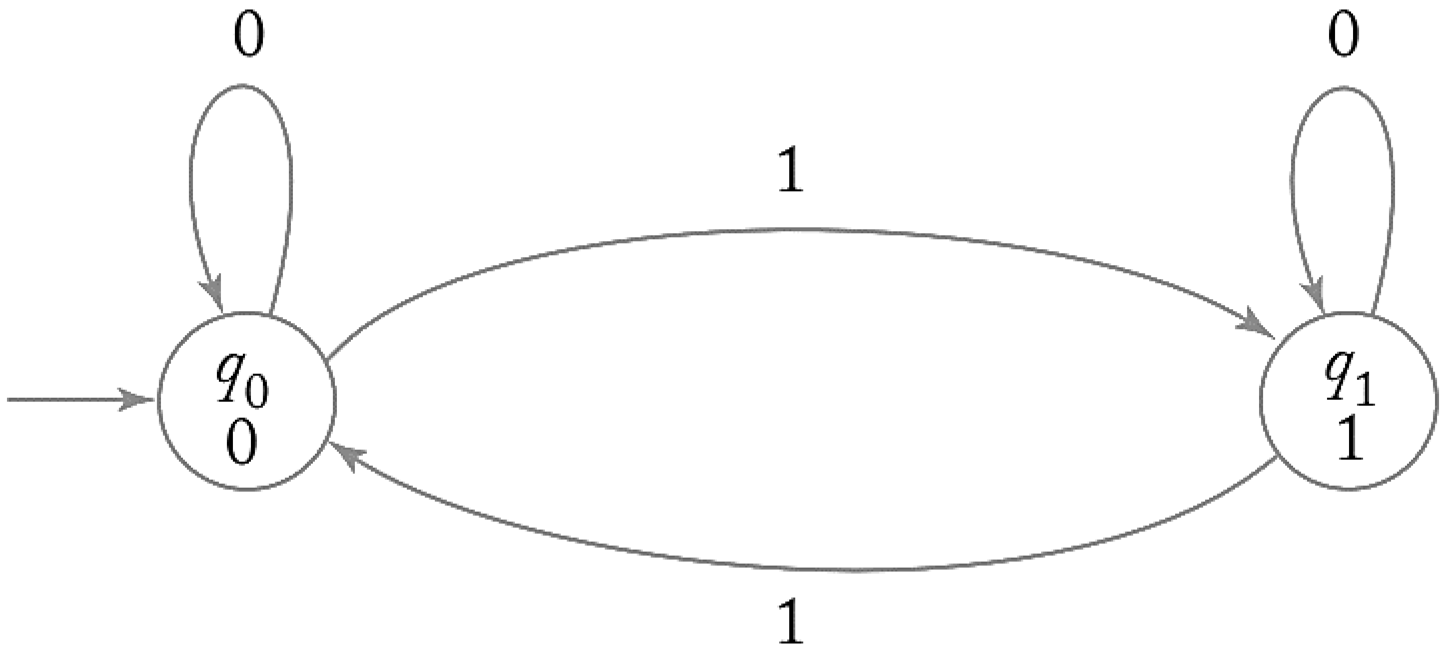
Moore Machine

- In a Moore machine every state is associated with an element of the output alphabet

- Whenever the state is entered, this output symbol is printed.

- A Moore machine is defined by the six tuple $M = (Q, \Sigma, \Gamma, \delta, \theta, q_0)$.
  - Q is a finite set of internal states,
  - Σ is the input alphabet,
  - Γ is the output alphabet,
  - $\delta : Q \times \Sigma \to Q$ is the transition function,
  - $\theta : Q \to \Gamma$ is the output function,
  - $q_0 \in Q$ is the initial state.

Moore Machine



Mealy to Moore Conversion

For different output associated with states, split them into separate states.

Present State	Next State (input = 0)	Output	Next State (input = 1)	Output
q1	q3	0	q2	0
q2	q1	1	q4	0
q3	q2	1	q1	1
q4	q4	1	q3	0

Present State	Next State (input = 0)	Output	Next State (input = 1)	Output

Mealy to Moore Conversion

For different output associated with states, split them into separate states.

Present State	Next State input = 0	Output	Next State input = 1	Output
q1	q3	0	q2	0
q2	q1	1	q4	0
q3	q2	1	q1	1
q4	q4	1	q3	0

Present State	Next State input = 0	Output	Next State input = 1	Output
q1	q3	0	q20	0
q20	q1	1	q40	0
q21	q1	1	q40	0
q3	q21	1	q1	1
q40	q41	1	q3	0
q41	q41	1	q3	0

Mealy to Moore Conversion

Merge the output columns to obtain the Moore machine. Output depends on present state only.

Present State	Next State	Output	Next State	Output
	input = 0		input = 1	
q1	q3	0	q20	0
q20	q1	1	q40	0
q21	q1	1	q40	0
q3	q21	1	q1	1
q40	q41	1	q3	0
q41	q41	1	q3	0

Present State	Next State	Next State	Output
	input = 0	input = 1	
q1	q3	q20	1
q20	q1	q40	0
q21	q1	q40	1
q3	q21	q1	0
q40	q41	q3	0
q41	q41	q3	1

Moore to Mealy machine

Separate output columns for each input value. Output is decided based on Next state only.

Present State	Next State		Output
	0	1	
q0	q3	q1	0
q1	q1	q2	1
q2	q2	q3	0
q3	q3	q0	0

Present State	Next State	Output	Next State	Output
	input = 0		input = 1	
q0				
q1				
q2				
q3				

Moore to Mealy machine

Separate output columns for each input value. Output is decided based on Next state only.

Present State	Next State		Output
	0	1	
q0	q3	q1	0
q1	q1	q2	1
q2	q2	q3	0
q3	q3	q0	0

Present State	Next State input = 0	Output	Next State input = 1	Output
q0	q3	0	q1	1
q1	q1	1	q2	0
q2	q2	0	q3	0
q3	q3	0	q0	0